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Systems and Methods for Detecting and Enabling Interpretation of Head and Hand Signals Using Body Language Informed Shape Scanning (BLISS)

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The Importance of Edge Computing in Computer Vision

- Frontier AI models generally assume that there's access to a large number of GPUs and an extensive computational infrastructure.
- Edge devices exploit compute platforms that are small and meant to be deployed in everyday environments.
- Edge-computing examples include home security cameras and autonomous vehicles.
  - These devices are typically marked by their efficient use of a smaller compute footprint with limited resources.
- In practical terms, no matter how small a CV compute platform is, there's always an application that will demand something smaller.
- Our work here shows how we can improve edge deployed CV systems to improve their application performance.

Introducing Body Language Informed Shape Scanning

- A patent entitled "System and Method for Detecting Gestures with Computer Vision" has been filed by inventors Lusine Kamikyan and Bruce Johnson.
- Our patent-pending algorithm is called Body Language Informed Shape Scanning (BLISS)
- It utilizes elastic shape analysis (ESA) to detect and track portions of a person's, robot's, or animal's body.
- This effort addresses the need for detecting and tracking an entity's appendages – be it a hand, head, tail, shoulders, etc.
- With this detection and tracking ability, we can now enable further applications that depend upon this knowledge.

How BLISS Works

1. Use NN-based object detector to find humans or animals
2. Perform contour extraction using multiple levels of sensitivity.
3. Use ESA to detect body portions meant to perform signaling
4. Select best representation of a part of the body using consensus among agents
5. Apply probability trace to track body part.

- Our algorithm is an application of neuro-symbolic computation.

BLISS in Operation

Potential Applications

Applications:

1. Signaling your robot butler
2. Assessing a game's referee's hand signals
3. Making interaction with robo-taxis more natural.
4. Enhancing an LLM’s ability to interact with people or animals.
5. Filtering out people detections in a camera.

We have presented a patent pending invention that enables the detection and tracking of bodily extremities for a given entity – however that's defined.

Questions?

Please reach out to johnson_bruce2@bah.com or Kamikyan_lusine@bah.com for more information